Problem 7

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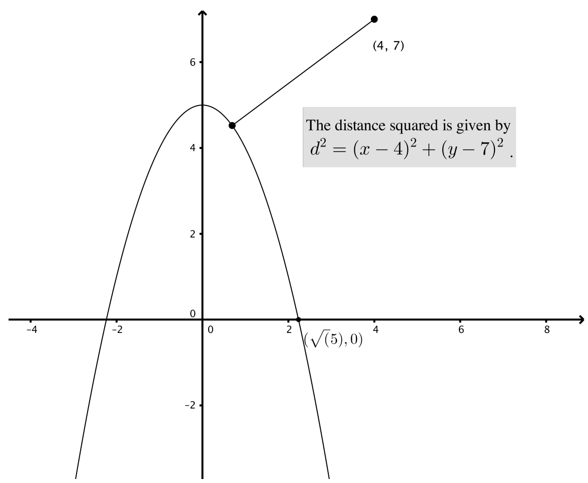
Problem 1 *What point on the parabola $y = 5 - x^2$ is closest to the point $(4, 7)$?*



Explanation.

We have "to minimize $d \iff$ to minimize d^2".

Objective function:

$$\begin{aligned} d^2 = (x-4)^2 + (y-7)^2 \text{ and } y = 5 - x^2 &\implies d^2 = (x-4)^2 + \left((5-x^2) - 7\right)^2 \\ &\implies f(x) = (x-4)^2 + (x^2+2)^2 \end{aligned}$$

We may assume $\text{Domain}(f) = [0, \sqrt{5}]$.

Critical points:

$$\begin{aligned} f'(x) &= 2(x-4) + 2(x^2+2) \cdot 2x \\ &= 4x^3 + 10x - 8. \end{aligned}$$

f' is defined everywhere on $(0, \sqrt{5}) \implies$ critical points only occur where $f'(x) = 0$:

$$\begin{aligned} f'(x) = 0 &\iff 4x^3 + 10x - 8 = 0 \\ \text{using a calculator to find an approximate value of } x &\implies x \approx 0.67628 \end{aligned}$$

Find global minimum:

$$\begin{aligned} f(0) &= 20 \\ f(0.67628) &\approx 17.1 \\ f(\sqrt{5}) &\approx 52.1 \\ &\implies \text{global minimum at } x \approx 0.67628 \end{aligned}$$